

Machine Learning 101: Lecture 3

Gradient Descent Optimization

Gradient descent is an iterative optimization algorithm used to find the parameters that minimize a loss function.

Parameter Update Rule

The parameter update formula is given by:

$$\theta = \theta - \alpha * \text{gradient}(J(\theta))$$

where α is the learning rate controlling the step size at each iteration.

If α is too large, the algorithm may overshoot the local minimum and diverge.

If α is too small, convergence will be excessively slow.

Python Implementation

```
```python
def gradient_descent(X, y, theta, alpha, num_iters):
 m = len(y)
 for _ in range(num_iters):
 grad = (1/m) * (X.T @ (X @ theta - y))
 theta = theta - alpha * grad
 return theta
```
```

| Parameter | Type | Purpose |
|-----------|--------|-------------------------|
| alpha | float | Learning rate step size |
| theta | vector | Model weight parameters |